



Introduction to Photogrammetry

Lecture 09

Digital Camera in Photogrammetry

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In Today's Class

1. Introduction
2. Advantages over Film-based Camera
3. Components of a Digital Camera
4. Image Formation

1. Introduction

- A digital camera is an electronic device that captures images and converts them into digital data. it is also called Solid State Camera.
- Unlike analog cameras, it stores images in numerical format, which makes them suitable for processing, analysis, and photogrammetric applications.
- Digital cameras play a fundamental role in photogrammetry because they provide the primary data source for extracting geometric and spatial information.
- The quality and accuracy of photogrammetric outputs depend heavily on camera characteristics.

2. Advantages over Film-based Camera

- The chief advantage of digital cameras over the classical film-based cameras is the instant availability of images for further processing and analysis. This is essential in real-time applications such as robotics, certain industrial applications, bio-mechanics, etc.
- Another advantage is the increased spectral flexibility of digital cameras.

2. Components of a Digital Camera

- A digital camera consists of several key components including the
 - lens,
 - image sensor,
 - shutter, and
 - processing unit.
- Each component contributes to capturing, controlling, and storing image data.

FRONT VIEW

SHUTTER TRIGGER

SHUTTER



CAMERA BODY

IMAGE SENSOR

www.photographyaxis.com

Image Sensors

- The image sensor is the most important component of a digital camera. It converts incoming light into electrical signals, which are then digitized into image data.

- **Types of Sensors (CCD vs CMOS)**
- Charge-Coupled Devices (CCD) and Complementary Metal-Oxide Semiconductor (CMOS) are the two main types of sensors.
- CCD sensors generally produce high-quality images, while CMOS sensors are more power-efficient and widely used.

CCD VS CMOS

Description		CCD	CMOS
01	Camera Components	Sensor + Optic Support Chips + Optics	Sensor + Optic: Support Chips Sometimes
02	Speed	Moderate to Fast	Fast
03	Sensitivity	High	Low
04	Noise	Low	Moderate
05	System Complexity	High	Low
06	Sensor Complexity	Low	High
07	Fill Factor	High	Low
08	Chip Output	Voltage (Analog)	Bits (Digital)
09	Pixel Signal	Electron	Voltage
10	Uniform Shuttering	High to Moderate	Low

Sensor Working Principle

- When light photons strike the sensor, they generate electrons proportional to the light intensity.
- These electrons are then converted into voltage and finally into digital values.

Spectral Sensitivity

- Digital sensors are sensitive to specific wavelength ranges.
- Most cameras capture visible light, but specialized cameras can capture infrared or multispectral data for remote sensing applications.

Radiometric Resolution

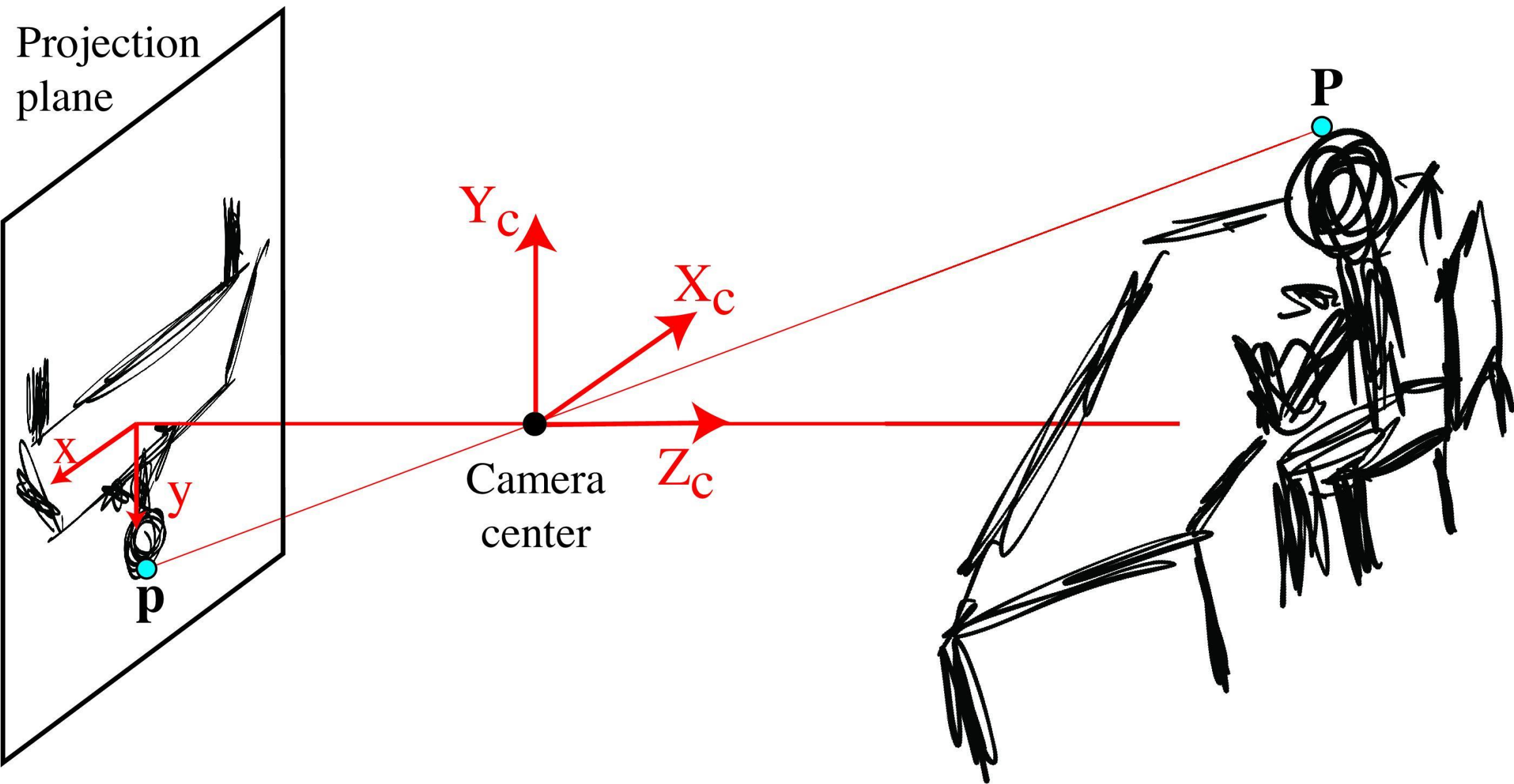
- Radiometric resolution refers to the sensor's ability to distinguish slight differences in energy. Higher radiometric resolution allows better detection of subtle variations in brightness

Spatial Resolution

- Spatial resolution refers to the ground area represented by each pixel. Higher spatial resolution results in more detailed images

4. Image Formation

- Image formation in a digital camera follows the central projection principle, where light rays from an object pass through the lens and converge onto the image sensor. This creates a scaled and inverted image.
- In photogrammetry, the central projection model assumes that all rays pass through a single point known as the perspective center. This concept is critical for geometric analysis and 3D reconstruction.



Color Image Formation

- Color images are formed using filters such as the Bayer filter, which assigns red, green, and blue values to pixels. These values are combined to produce a full-color image.

pixels

- A digital image is composed of pixels, which are small picture elements representing intensity values.
- The spatial resolution of an image depends on the number and size of these pixels.

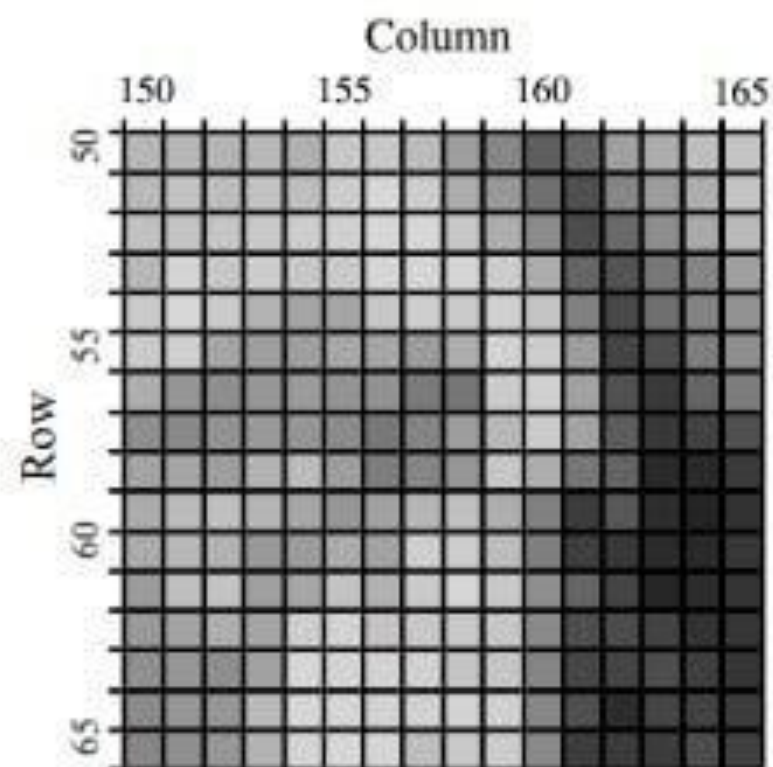
Grayscale

223	150	91
39	71	150
221	150	221

RGB

255	79	42			
79	255	42	181		
42	79	255	102	63	
	102	255	29	105	
		63	105	255	

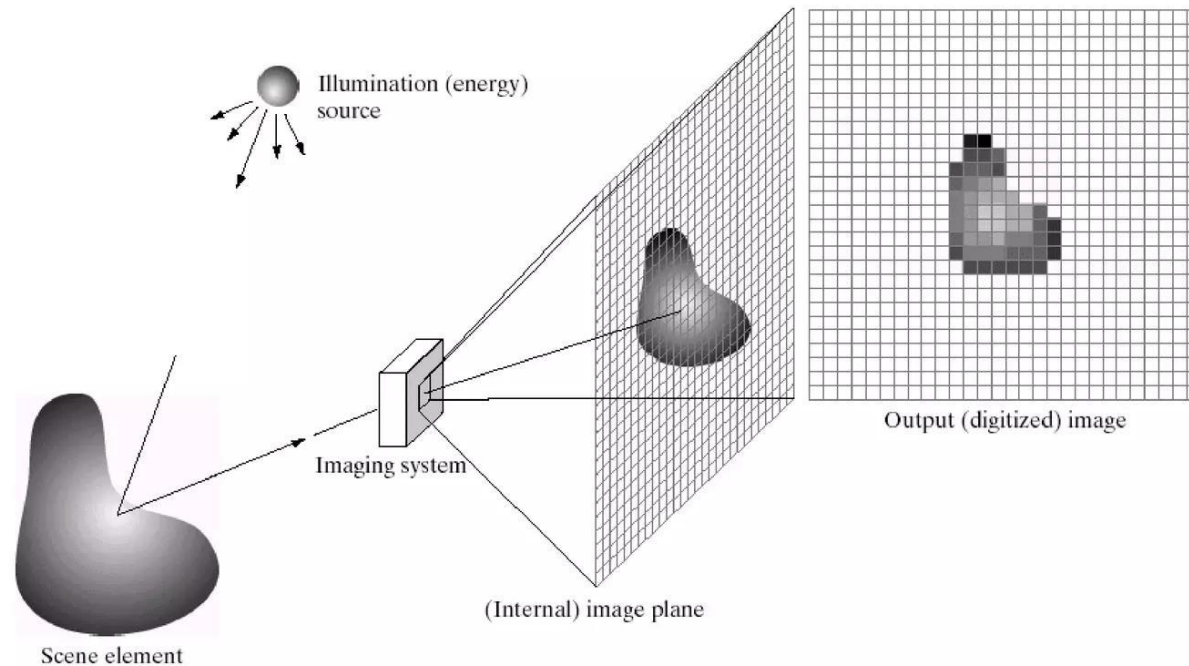
$$\begin{bmatrix} 0 & 128 & 255 \\ 64 & 192 & 32 \\ 255 & 100 & 50 \end{bmatrix}$$



		Column															
		150				155				160				165			
Row	50	183	183	181	184	177	200	200	189	159	135	94	105	160	174	191	196
		186	195	190	195	191	205	216	206	174	153	112	80	134	157	174	196
55		194	196	198	201	206	209	215	216	199	175	140	77	106	142	170	186
		184	212	200	204	201	202	214	214	214	205	173	102	84	120	134	159
60		202	215	203	179	165	165	199	207	202	208	197	129	73	112	131	146
		203	208	166	159	160	168	166	157	174	211	204	158	69	79	127	143
65		174	149	143	151	156	148	146	123	118	203	208	162	81	58	101	125
		143	137	147	153	150	140	121	133	157	184	203	164	94	56	66	80
		164	165	159	179	188	159	126	134	150	199	174	119	100	41	41	58
		173	187	193	181	167	151	162	182	192	175	129	60	88	47	37	50
		172	184	179	153	158	172	163	207	205	188	127	63	56	43	42	55
		156	191	196	159	167	195	178	203	214	201	143	101	69	38	44	52
		154	163	175	165	207	211	197	201	201	199	138	79	76	67	51	53
		144	150	143	162	215	212	211	209	197	198	133	71	69	77	63	53
		140	151	150	185	215	214	210	210	211	209	135	80	45	69	66	60
	65	135	143	151	179	213	216	214	191	201	205	138	61	59	61	77	63

What is a Digital Image?

- A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels





Thank you!

Any Questions
